PROCUREMENT PACKAGE:	EL001 – Switchrooms
SWITCHROOM NUMBER/S	**EQUIPMENT NAME**
F00-SR-001	33kV Switchroom
F12-SR-001 AND F12-SR-002	Primary Crushing 11kV Switchroom and 415V Switchroom
F13-SR-001	Sag Mill 415V Switchroom
F13-SR-002	Ball Mill 415V Switchroom
F13-SR-003	Sag and Ball Mill VSD Switchroom
F13-SR-004	Recycle Crushing 415V Switchroom
F13-SR-005	Mill Chillers & Reagents 415V Switchroom
F16-SR-001	Flotation 11kV Switchroom
F16-SR-002	Flotation, Blowers & Compressors 415V Switchroom
F21-SR-001 (SP2)	CIL4 415V Switchroom
F22-SR-001 (SP2)	Carbon Treatment & Elution 415V Switchroom
F23-SR-001 AND F23-SR-002	Final Tails Pumps 11kV Switchroom and 415V Switchroom
F28-SR-001 AND F28-SR-002	Ultrafine Grinding 11kV Switchroom and 415V Switchroom
F72-SR-001 (SP2)	Reagents 415V Switchroom

SWITCHROOM DESCRIPTION:

1. **33kV, 11kV and 415V transportable type switchrooms comprising:**

- Transportable building – standalone structure mounted on a structural steel base frame with supporting columns, landings (at equipment entry and personnel entry ends) and stairs (except for switchrooms F13-SR-001, F13-SR-002 and F13-SR-003 that will be installed into steel structure on site);
- Switchroom buildings may be comprised as a combination of the following (refer to switchroom layout drawings);
    - HV switchroom – 33kV standalone
    - 11kV switchroom
    - Airlock
    - LV switchroom
    - Transformer room (separate enclosures for dry transformers)
- Airlocks shall house Remote Switching Panel(s) for safe operation of 33kV and 11kV switchgears (all panels)
- Airlocks shall house NHP TemRack IR Local Control Station(s) for safe operation of LV Air Circuit Breaker incomers.
- Shrouded Emergency Shutdown Pushbutton as noted on the drawings shall be located inside the airlock. Emergency Shutdown shall trip all incomers and upstream feeder circuit breakers (Field wiring by others). Emergency shutdown shall also trip automatic transfer switch.
- Fit for purpose air-conditioning - Ducted, Split systems or Computer Room Air Conditioning (CRAC), appropriate for the heat loads with N+1 redundancy. The redundant unit shall maintain switchroom temperatures within limits when one unit fails.
- Fire detection and suppression systems comprising:
    - Smoke and Heat Detectors
    - Manual call points
    - VESDA system
    - Fire Indicating Panel
    - Inergen (or equivalent) gas suppression system inclusive all pipe works. Gas canisters shall be located indoors
    - CO_2 fire extinguishers inside and outside as per National Construction Code (NCC) Requirements
    - All signages.
    - VESDA system shall sample air from the following:
        - Switchroom Ceiling – HV Room, LV Room and Transformer Room/s
        - HV switchgear (LV compartment only)
        - Incomer and Cable zones of each tier in LV MCCs
        - LV VSDs > 315kW
        - Communication panel
        - DCS and RIO panel
- Switchroom LV DBs, Lighting (incl. emergency lights), GPOs and RJ45 outlets, defibrillator, rescue kits and CPR signage;
- All doors leading from the airlock into switchrooms shall have a 200 x 200mm laminated safety glass window to allow operators to see inside the room for fire/smoke prior to entry.

2. **Switchrooms will house the following electrical equipment as per the electrical equipment lists and as follows:**

- 33kV, 11kV and 415V switchgear, Dry type transformers, HV and LV VSD's, 240V AC UPS & batteries, 48V DC battery charger and batteries, Distribution boards, Fire panels, HVAC equipment, DCS panel, Switchroom IO and Communications cabinets
- Free-issued LV dry type transformers (IP00) shall be housed inside a purpose-built transformer room/enclosure with air conditioning (N+1)
- Transformer rooms/enclosures shall have a removable wall panel (or double doors) for transformer change out in case of failure. If a removable panel is used it shall be fitted with a personnel access door. All access door(s) shall be equipped with a Fortress key lock to ensure supply power is isolated and earthed prior to personnel entry - Fortress keys to be fitted on supply feeder earth switch and keyed alike to Transformer room personnel access doors
- Transformer installation and removal into and out of the room shall be facilitated with C-channels installed into/onto the transformer enclosure floor. The dimensions and spacing of the C-channels shall be determined by the wheel and spacing dimensions of the dry transformers. C-channels shall be extended onto the external platform allowing the transformer to be pushed into the room or pulled out of the room if required

3. Switchroom tender special requirements

- The switchroom builder will supply a drawing of the switchroom builders' standard switchgear drop box that the builder has supplied on previous switchroom projects for evaluation purposes.
- LV MCC arc fault detection current transformers to be installed in the transformer rooms on purposed build frame on the transformer's secondary.

STANDARDS:

AS 1170 – Structural Design Actions
AS 1670 - Fire detection, warning, control and intercom systems - System design, installation and commissioning
AS 2067 – Substations and High Voltage Installations Exceeding 1kV AC
AS 2293 – Emergency Escape Lighting and Exit Signs for Buildings
AS 2676 – Guide to the Installation, Maintenance, Testing and Replacement of Secondary Batteries in Buildings
AS 3000 – Electrical Installations (known as the Australian Wiring Rules)
AS 4100 – Steel Structures

REFERENCE DOCUMENTS AND DRAWINGS:

EQUIPMENT LISTS	
Document Number	**Description**
2000-F00-LST-EL-11032	F00-SR-001 33kV Switchroom Equipment List and Cable Schedule
2000-F12-LST-EL-11040	F12-SR-001 Primary Crushing 11kV Switchroom Equipment List and Cable Schedule
2000-F12-LST-EL-11044	F12-SR-002 Primary Crushing 415V Switchroom Equipment List and Cable Schedule
2000-F13-LST-EL-11036	F13-SR-001 SAG Mill 415V Switchroom Equipment List and Cable Schedule
2000-F13-LST-EL-11033	F13-SR-002 Ball Mill 415V Switchroom Equipment List and Cable Schedule
2000-F13-LST-EL-11041	F13-SR-003 SAG and Ball Mill VSD Switchroom Equipment List and Cable Schedule
2000-F13-LST-EL-11087	F13-SR-004 Recycle Crushing 415V Switchroom Equipment List and Cable Schedule
2000-F13-LST-EL-11039	F13-SR-005 Mill Chillers 415V Switchroom Equipment List and Cable Schedule
2000-F16-LST-EL-11035	F16-SR-001 Flotation 11kV Switchroom Equipment List and Cable Schedule
2000-F16-LST-EL-11090	F16-SR-002 Flotation, Blowers & Compressors 415V Switchroom Equipment List and Cable Schedule
2000-F21-LST-EL-12001	F21-SR-001 CIL4 415V Switchroom Equipment List and Cable Schedule (SP2)
2000-F22-LST-EL-12001	F22-SR-001 Carbon Treatment & Elution 415V Switchroom Equipment List and Cable Schedule (SP2)
2000-F23-LST-EL-11037	F23-SR-001 Final Tails Pumps 11kV Switchroom Equipment List and Cable Schedule
2000-F23-LST-EL-11038	F23-SR-002 Final Tails Pumps 415V Switchroom Equipment List and Cable Schedule
2000-F28-LST-EL-11088	F28-SR-001 Ultrafine Grinding 11kV Switchroom Equipment List and Cable Schedule
2000-F28-LST-EL-11034	F28-SR-002 Ultrafine Grinding 415kV Switchroom Equipment List and Cable Schedule
2000-F72-LST-EL-11043	F72-SR-001 Reagents 415V Switchroom Equipment List and Cable Schedule
SWITCHROOM LAYOUTS	
Document Number	**Description**
2000-F00-DRG-EL-10001	F00-SR-001 33kV Switchroom Equipment Layout
2000-F12-DRG-EL-10001	F12-SR-001 & 002 Primary Crushing 11kV and 415V Switchroom Equipment Layout
2000-F13-DRG-EL-10101	F13-SR-001 SAG Mill 415V Switchroom Equipment Layout
2000-F13-DRG-EL-10102	F13-SR-002 Ball Mill 415V Switchroom Equipment Layout
2000-F13-DRG-EL-10103	F13-SR-003 SAG and Ball Mill VSD Switchroom Equipment Layout
2000-F13-DRG-EL-10106	F13-SR-004 Recycle Crushing 415V Switchroom Equipment Layout
2000-F13-DRG-EL-10121	F13-SR-005 Mill Chillers & Reagents 415V Switchroom Equipment Layout
2000-F16-DRG-EL-10001	F16-SR-001 Flotation 11kV Switchroom Equipment Layout
2000-F16-DRG-EL-10101	F16-SR-002 Flotation 415V Switchroom Equipment Layout
2000-F21-DRG-EL-10101	F21-SR-001 CIL4 415V Switchroom Equipment Layout (SP2)
2000-F22-DRG-EL-10101	F22-SR-001 Carbon Treatment & Elution 415V Switchroom Equipment Layout (SP2)
2000-F23-DRG-EL-10001	F23-SR-001 & 002 Final Tails Pump 11kV and 415V Switchroom Equipment Layout
2000-F28-DRG-EL-10001	F28-SR-001 & 002 Ultrafine Griding 11kV and 415V Switchroom Equipment Layout
2000-F72-DRG-EL-10001	F72-SR-001 Reagents 415V Switchroom Equipment Layout
BLOCK DIAGRAMS	

Document Number	Description
2000-F12-BLK-IC-10001	F12-SR-001 & 002 Primary Crushing Switchroom Communications Block Diagram
2000-F13-BLK-IC-10301	F13-SR-001 SAG Mill Switchroom Communications Block Diagram
2000-F13-BLK-IC-10311	F13-SR-002 Ball Mill Switchroom Communications Block Diagram
2000-F13-BLK-IC-10321	F13-SR-003 SAG and Ball Mill VSD Switchroom Communications Block Diagram
2000-F13-BLK-IC-10341	F13-SR-005 Mill Chillers & Reagents Switchroom Communications Block Diagram
2000-F13-BLK-IC-10401	F13-SR-004 Recycle Crushing Switchroom Communications Block Diagram
2000-F16-BLK-IC-10001	F16-SR-001 & 002 Flotation Switchroom Communications Block Diagram
2000-F21-BLK-IC-10001	F21-SR-001 CIL4 Switchroom Communications Block Diagram (SP2)
2000-F22-BLK-IC-10001	F22-SR-001 Carbon Treatment & Elution Switchroom Communications Block Diagram (SP2)
2000-F23-BLK-IC-10001	F23-SR-001 Final Tails Pump Switchroom Communications Block Diagram
2000-F28-BLK-IC-10001	F28-SR-001 Ultrafine Griding Switchroom Communications Block Diagram
2000-F72-BLK-IC-10001	F72-SR-001 Reagents 415V Switchroom Communications Block Diagram
SINGLE LINE DIAGRAMS	
Drawings Number	**Description**
2000-F00-SLD-EL-10001	General FIM Site, Overall Sitewide 33kV Distribution, Single Line Diagram, Sheet 1
2000-F00-SLD-EL-10011	General FIM Site, Overall Sitewide 33kV Distribution, Single Line Diagram, Sheet 2
2000-F00-SLD-EL-10012	General FIM Site, Overall Sitewide 33kV Distribution, Single Line Diagram, Sheet 3
2000-F00-SLD-EL-10002	F00-SW-010 33kV Switchboard, Single Line Diagram, Sheet 1
2000-F00-SLD-EL-10003	F00-SW-010 33kV Switchboard, Single Line Diagram, Sheet 2
2000-F00-SLD-EL-10004	F00-SW-010 33kV Switchboard, Single Line Diagram, Sheet 3
2000-F00-SLD-EL-10005	F00-SW-010 33kV Switchboard, Single Line Diagram, Sheet 4
2000-F00-SLD-EL-10006	F00-SW-010 33kV Switchboard, Single Line Diagram, Sheet 5
2000-F12-SLD-EL-10001	Primary Crushing, F12-SW-012 11kV Switchboard, Single Line Diagram
2000-F16-SLD-EL-10001	Flotation, F16-SW-011 11kV Switchboard, Single Line Diagram, Sheet 1
2000-F16-SLD-EL-10002	Flotation, F16-SW-011 11kV Switchboard, Single Line Diagram, Sheet 2
2000-F16-SLD-EL-10003	Flotation, F16-SW-011 11kV Switchboard, Single Line Diagram, Sheet 3
2000-F23-SLD-EL-10001	Final Tails Pumps, F23-SW-016 11kV Switchboard, Single Line Diagram, Sheet 1
2000-F23-SLD-EL-10002	Final Tails Pumps, F23-SW-016 11kV Switchboard, Single Line Diagram, Sheet 2
2000-F28-SLD-EL-10001	Ultrafine Grinding, F28-SW-013 11kV Switchboard, Single line Diagram, Sheet 1

REFERENCE SPECIFICATIONS:

1. 2000-F00-STS-EL-10004_3 – High Voltage Electrical Testing Specification
2. 2000-F00-STS-EL-10007_0 – Low Voltage Distribution Board Specification
3. 2000-F00-STS-EL-10008_0 – Electrical Enclosures Specification
4. 2000-F00-STS-EL-10009_0 – Cables Specification
5. 2000-F00-STS-EL-10011_0 – Switchrooms Specification
6. 2000-F00-STS-EL-10014_2 – Preferred Equipment Suppliers for Electrical, Instrumentation And Controls Specification
7. 2000-F00-STS-GE-10001_3 – Packing and Shipping Specification
8. 2000-F00-STS-GE-10002_1 – Site Specific Information
9. 2000-F00-STS-GE-10003_3 – Plant Numbering Specification
10. 2000-F00-STS-ST-10001_3 – Steelwork Fabrication
11. 2000-F00-STS-ST-10002_2 – Steelwork Erection Specification
12. 2000-EL01A-B01-10001 – Vendor Document Schedule
13. SPEC-00-G-05_12 – Technical Specification for Protective Coatings

SCOPE OF WORK

The vendor shall design, fabricate, assemble, test and transport (to site) the transportable switchrooms, including installation, wiring, terminating, and testing as far as reasonably practical, all equipment listed on the switchroom equipment lists (including all free-issued equipment), prior to the switchrooms being shipped to site.

Work Included

- Design, fabrication, fit-out, and testing of fourteen (14) transportable switchroom buildings, over two separable portions (SP1 and SP2), including, but not limited to:
  - Paint system CS1 (or equivalent), structurally certified base frames, stairs;
  - Paint system CS1 (or equivalent), structurally certified, equipment platforms (rated for a minimum of 10kPA) and switchroom support columns;

- Switchroom building/s;
- Design, supply and installation of one Remote Switching Panel per switchboard to be mounted inside the airlock for switching/control of all 33kV and 11kV circuit breakers and fused contactors, complete with switches and indication semaphores (based on the Single Line Diagrams);
- Supply and installation of one NHP TemRack IR LCS per LV MCC Incomer;
- Normal, emergency and external lighting;
- General power and Network (RJ45) outlets;
- Air conditioning, complete and operational (to be tested during switchroom FAT) with piping, drains and isolators, to be installed such that it can remain attached during transport (if practical);
- Air conditioner controller, one per room (Including transformer enclosure), with integral temperature sensing. To be wired directly to the Air conditioner units.
- Room combined temperature and humidity transmitter, one per room. Hardwired analogue output to DCS;
- Lighting & small power and services (HVAC) 415V distribution board;
- HV VSD Auxiliary Supplies 415V AC Distribution board located in F16-SR-001;
- Vendor supplied 240V AC UPS, Battery Cabinet and Batteries, and distribution board. Note that batteries shall be delivered separately to site at a later date (As advised by Principal). Allowance for one smaller set of batteries for FAT at vendor premises to be allowed for;
- Vendor supplied Communications cabinet and fit out in accordance with Switchroom equipment and cable lists;
- Vendor supplied Active Harmonic Filters as specified, including inter-cabling and CTs;
- Fire detection equipment, fire indication panel and VESDA;
- Gas suppression systems incl. Inergen (or equivalent) gas cylinder/s, valves, piping and cabinets;
- Supply and Installation of 415V MCC's per switchroom, including 415V Emergency and Essential switchboards;
- Supply and installation of Incomer ACB Arc Flash CT's to be mounted on LV side of the 415V transformers;
- Installation of free-issue equipment (refer to switchroom equipment lists);
- Installation of free-issued HV Switchgear Core balance CTs – to be mounted by Vendor in the drop box of the associated switchgear tier.
- Vendor supplied 48VDC Battery Charger and batteries - note that batteries shall be delivered separately to site at a later date (As advised by Principal). Allowance for one smaller set of 48V DC batteries for FAT at vendor premises to be allowed for;
- Allowance for any ducting and penetrations as required by equipment OEMs;
- Conduit, cable tray and cabling between all equipment installed in or supplied with the room; or inter wiring between equipment located in combined switch rooms F12-SR-001 & F12-SR-002, F23-SR-001 & F23-SR-002 and F28-SR-001 & F28-SR-002;
- HV and LV Cabling as specified on the Switchroom cable schedules between HV Switchgear/LV MCC and VSDs;
- Engagement of equipment OEM for support for equipment installation and testing. Specific engagement shall be sought with ABB for HV Switchgear CT installation and function testing of interlock systems;
- Earthing of equipment in the room to the switchroom earth bar;
- Equipment nameplates and labels;
- Cable labels;
- Signages;
- Switchroom landings, support structure and stairs (except for above stated exceptions);
- Installation of key exchange boxes where specified in the SLD's;
- Any other items which are normally supplied with this equipment.

- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- Factory Acceptance Testing at manufacturer's premises including CT / relay injection testing;
- Supply of all necessary documentation, calculations where applicable, drawings and test documentation;
- Suitable internal bracing for road transport to site Owner FIM
- Spare parts list to be provided;
- Unloading and storage of free-issued equipment. Acceptance of free-issued equipment from September 2024;
- Transportation of all switchrooms, HVAC units (if separate), landings & stairways in compliance with Main Roads regulations to Owner FIM

Special Notice

- The switchrooms package will be awarded together with the LV MCC's package EL08 to the successful tenderer.

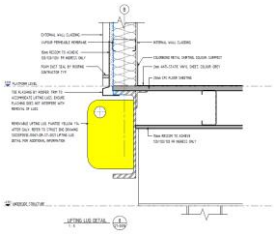
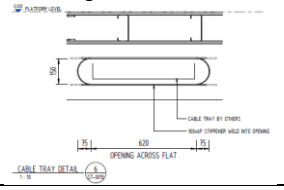
Work Excluded

- Fabrication or design of free-issue items;
- Offloading the equipment at site;
- Installation and commissioning beyond support specified above.

SWITCHBOARDS AND TRANSFORMERS PER ROOM							
NO.	BUILDING NO.	SWITCHROOM NUMBER	EQUIPMENT NAME	33KV SWITCHBOARD	11KV SWITCHBOARDS	415V LV MCCS	415V DRY TRANSFORMERS
1	Building 1 (SP1)	F00-SR-001	33kV Switchroom	F00-SW-001	NA	NA	NA
2	Building 2 (SP1)	F12-SR-001	Primary Crushing 11kV Switchroom	NA	F12-SW-012	NA	NA
3		F12-SR-002	Primary Crushing 415V Switchroom	NA	NA	F12-SW-001 F12-SW-101 (Essential)	F12-TR-001 (2500kVA)
4	Building 3 (SP1)	F13-SR-001	Sag Mill 415V Switchroom	NA	NA	F13-SW-002 F13-SW-102 (Essential)	F13-TR-002 (2500kVA)
5	Building 4 (SP1)	F13-SR-002	Ball Mill 415V Switchroom	NA	NA	F13-SW-003 F13-SW-103 (Essential)	F13-TR-003 (2500kVA)
6	Building 5(SP1)	F13-SR-003	Sag and Ball Mill VSD Switchroom	NA	NA	NA	N/A
7	Building 6 (SP1)	F13-SR-004	Recycle Crushing 415V Switchroom	NA	NA	F13-SW-014 F13-SW-114 (Essential)	F13-TR-014 (2500kVA)
8	Building 7 (SP1)	F13-SR-005	Mill Chillers & Reagents 415V Switchroom	NA	NA	F13-SW-020 F13-SW-120 (Essential)	F13-TR-020 (2500kVA)
9	Building 8 (SP1)	F16-SR-001	Flotation 11kV Switchroom	NA	F16-SW-011	NA	NA
10	Building 9 (SP1)	F16-SR-002	Flotation, Blowers & Compressors 415V Switchroom	NA	NA	F16-SW-004 F16-SW-005 F16-SW-008 (Emergency) F16-SW-104 (Essential)	F16-TR-004 (2500kVA) F16-TR-005 (2500kVA) F16-TR-008 (630kVA)
11	Building 10 (SP2)	F21-SR-001	CIL4 415V Switchroom	NA	NA	F21-SW-006 ~~F21-SW-106~~ ~~(Essential)~~ F21-SW-007 ~~F21-SW-107~~ ~~(Emergency)~~	F21-TR-006 (2500kVA) F21-TR-007 (2500kVA)
12	Building 11 (SP2)	F22-SR-001	Carbon Treatment & Elution 415V Switchroom (SP2)	NA	NA	F22-SW-030 F22-SW-031 F22-SW-131 (Emergency)	F22-TR-030 (2500kVA) F22-TR-031 (2500kVA)
13	Building 12 (SP1)	F23-SR-001	Final Tails Pumps 11kV Switchroom	NA	F23-SW-016	NA	NA
14		F23-SR-002	Final Tails Pumps 415V Switchroom	NA	NA	F23-SW-023 F23-SW-123 (Emergency)	F23-TR-023 (1000kVA)
15	Building 13 (SP1)	F28-SR-001	Ultrafine Grinding 11kV switchroom	NA	F28-SW-013	NA	NA
16		F28-SR-002	Ultrafine Grinding 415V switchroom	NA	NA	F28-SW-009 F28-SW-109 (Essential)	F28-TR-009 (1000kVA)
17	Building 14 (SP2)	F72-SR-001	Reagents 415V Switchroom	N/A	N/A	F72-SW-072 F72-SW-172 (Emergency)	F72-TR-072 (1600kVA)

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
General				
Type of Building	-	Transportable Switchroom		1
Method of Installation	-	Above Ground, Bottom Entry		1
Codes	-	Building Code of Australia (BCA) and FESA		1
Compliance with Applicable Australian Standards	-	Noted above		1
Equipment Layout Drawing	-	Refer to RFQ pack – sketches are Indicative only, vendor to provide construction layout for review/approval	Noted	3, 4
External Layout Drawing	-	By vendor	Yes	3, 4
Penetrations Drawing	-	By vendor, allow for dropboxes and penetrations below HV switchboards, 415V MCCs, HV and LV VSDs, UPS, DBs, Comms and other cabinets	Yes	1, 4
Manufacturer	-		POWINS	2
Country of Manufacture	-		Australia	2
Environmental Conditions				
Plant Location	-	Western Australia		1
Service Location	Indoor/Outdoor	Outdoor, direct sunlight and rain		1
Ambient Air Temperature – Min	°C	-5		1
Ambient Air Temperature – Max	°C	50		1
Humidity	%	0 to 99		1
Altitude	M	< 1000m		1
Other Data	-	2000-F00-STS-GE-10002 - Site Specific Information		1
Structural Design Criteria				
Wind Loading	-	AS 1170.2		1
Wind Region	-	A0		1
Importance Level	-	2		1
Wind Terrain Category	-	2.5		1
Annual Probability of Exceedance (ultimate)	-	1/200		1
Earthquake Loading	-	AS 1170.4		1
Sub-soil Class	-	Class Be		1
Hazard Factor, Z	-	0.09		1
Probability Factor, k_p	-	0.75		1
Annual Probability of Exceedance (ultimate)	-	1/250		1
Support Reactions	-	To be determined in detailed design.	Noted	2, 6
Construction				
Building Steel Base	-	Rigid Self Supporting Base Frame with certified lifting facilities, Subfloor cable ladders to be located within baseframe. Surface coating - paint system CS1 or equivalent	Yes. Surface coating CS1 to structural steel sections only. "C" Purlins, Wall & Roof Portal frame, equipment mounting frames, walkmesh, stair treads to be HDG. Drop boxes to be powercoated	1, 7
Lifting Lugs	-	Yes – stamped with safe working load	Yes	1
Lifting	-	Switchroom shall be provided with lifting diagram and shall be sufficiently rigid to lift with a single spreader bar.	Lifting diagram is provided , however the rooms are designed for multiple point lift down each side of the room	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Building Material for Walls and Roof	-	Welded steel stud and structural portal frames, insulation in cavities, with colorbond internal (shallow profile) and external cladding (trim deck profile)	Powins bolted construction method only. COLORBOND® Steel - TRIMDEK® or approved equivalent	3
Materials & Thickness of Insulation	VTA to meet temperature requirements, minimum as follows below:			1
Wall	mm	Minimum R3.0 Rockwool Insulation Batts	Yes	3
Ceiling	mm	Minimum R3.0 Rockwool Insulation Batts	Yes	3
Moisture barrier	-	Yes VTA provide detail	Sarking foil on portal frame	3
Roof Slope Orientation	-	VTA	Skillion 5 degrees	3
Roof Gutters and Down Pipes	-	No	No	1
Flooring	-	Compressed Fibre cement (Minimum thickness of 18mm) with non-flammable industrial grey vinyl (Minimum thickness of 2mm) with black skirting.	Noted and we comply	1
Windows	-	Access/Exit doors to be equipped with a minimum 200mm by 200mm laminated safety glass window.	Noted and we comply	1
Air Lock Room	-	Yes, air lock entry for personnel access door.	Noted and we comply	1
Switchroom doors – PERSONNEL ACCESS TO AIRLOCK	-	One Personnel access door with laminated 200x200mm safety glass window for each room leading off airlock.	Noted and we comply	1
Switchroom doors - EQUIPMENT ACCESS	-	One two-leaf equipment door onto equipment deck. Door swing to both be 170 degrees with door catch to allow maximum deck space when manoeuvring equipment on deck.	Noted and we comply	1
Transformer room – General	-	Structurally designed to suit the dry type transformer weight (up to 7000kg); Transformer to be permanently fixed with wheels removed and securely stored within the transformer room.	Noted.	1
Transformer room door – 415V TRANSFORMER ROOM ACCESS	-	One Personnel access door WITH Fortress key lock – keyed alike to upstream HV circuit breaker earth switch. Include removable wall panels and rails for Tx removal	Noted and we comply	1
Transformer room - Gland and kit plates	-	Gland plate and kick plate to be mounted on floor under both the HV and LV sides of the transformer.	Noted and we comply	1
Transformer room - C Channel		C-channel to be mounted in the floor, such that the top of the channel sits flush with the floor. Channel shall be mounted the same width as shown on Transformer GA.	Noted	1
Material for Internal Walls	-	Non-Combustible	Yes (Stramit Minirib)	3
Internal ceiling and wall finishing colour	-	Surfmist or equivalent	Surfmist	3

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Internal Floor colour	-	VTA – Grey preferred	Noted	3
External painting compliant with specification surface protection system CS1	Yes / No	Yes	Yes Applies to structural steel , base frame, platforms, stumps stair stringers, etc. Excluded “C”, “Z” and in subfloor and walls, and portal wall frames and heavy equipment mounting frames which are all HDG.)	1
Switchroom external colour	-	Lysaght Shale Grey	Noted and we comply	1
Base frame member surface protection system CS1	-	Shale Grey or equivalent	Noted and we comply	2
Dimension from bottom of subframe to finished floor level	mm	VTA	~520mm 	2
Cable Ladder (mounted under floor inside baseframe members)		Stainless Steel 316 Cable Ladder	Yes (600 W)	1
Support Structure				
Supply of support structure	Yes / No	Yes, building columns/legs will be 1.8m high from the ground to the underside of the building steel base frame. Support structure to include cantilever supports for 2 laterally (side by side), 3 rows high (vertically), 600mm cable ladders to run on both sides under the switchroom for field wiring. Cable ladders will be supplied by others. Surface coating - paint system CS1 or equivalent	Yes 3 x 1300mmL (75x75x8 EA) cable ladder support down the long side stumps only	1
Supply of stairs	Yes / No	Yes	Yes	1
Supply of landings	Yes / No	Yes. Include C Channels/rails embedded flush to landing where applicable outside of TX room for install/removal purposes of TX	Yes Supply of removal Installation Channels/rails for install/removal purposes of TX (one set provided for project)	1
Provided with cable ladder supports for subfloor	Yes / No	Yes	Yes, through subfloor beams 	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Bolted for ease of transport and assembly on site	Yes / No	Yes	Yes	1
Landings to be attached to switchroom for site transport	Yes / No	Yes	Yes. Mostly achievable F13-SR-004, F16-SR-001 & F16-SR-002, the 2x end platforms need to be removed for transport. These will be shipped separately to site, for site reassembly and the reconnection of the air conditioner gas pipe work & electrical connections. This reassembly and reconnection are by others	1
Support structure/legs, landings and stairs surface coating	-	Paint system CS1 or equivalent	Yes	1
Support structure paint colour	-	Shale Grey or equivalent	Noted and we comply	1
External Cable Entry				
Cable entry	-	Bottom Entry via Flooring	Yes	1
Cable drop boxes provided	-	Required under HV Switchgear, 415V MCCs, HV and floor mounted LV VSDs, AHF's and all floor mounted equipment. Constructed with removable side and bottom gland plates. To be within shipping envelope of building and remain installed during transport. VTA to provide detail drawings of offer. Surface coating - paint system CS1 or equivalent	Vendor Typical Drop Box design to be utilised. Drop boxes to be powder coated	3
Floor penetrations for cable entry	-	As above and locate to avoid floor beams and joists, Sealed. Where gland plate is exposed on finished floor surface, kick plate shall be installed. Eg. Under wall mounted DB's	Noted and we comply	1
Floor penetration gland plates	-	5mm Aluminium, earthed with neoprene gasket	Noted and we comply	1
Mechanical Services Requirements				
Internal Room Temperature to be maintained	Degrees °C	27 °C Maximum	Noted	1
Fouling Factor to be applied	%	20	Noted	1
Airconditioning Design requirements	-	Air conditioners, complete with piping and isolator, as required to maintain internal ambient temperature and provide N+1 redundancy with 20% fouling factor	Noted and we comply	1
Split system make and model	-	VTA		2
Air conditioner types - HV VSD rooms	-	Multiple Split Systems (N+1) *preferred* – otherwise, high capacity commercial ducted or CRAC systems installed on equipment platforms. (Non-preferred are separate ground mounted units).	Multiple Split Systems offered (see commercial offer for details)	3

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Air conditioning types – All other (non-HV VSD) rooms	-	Multiple Split Systems (N+1) – single phase, wall mount or 3 phase, in ceiling, high kW commercial type units with individual isolators	Ceiling cassette types offered (see commercial offer for details)	3
Air conditioner internal condensation drip trays (splits)	-	To be included under each unit, (on wall or ceiling **where relevant**)	Ceiling Cassettes have built in drip drays	1
Air conditioner temperature control	-	Single wall mounted controller to cover all split systems within a room space.	Noted and we comply	1
Air conditioner controller make/model			Mitsubishi PAR-U02MEDA ME	2
Air conditioner controller - Communication		Modbus TCP capable	Noted	3
Air conditioner controller status contacts to DCS	-	Yes, 2 relay outputs to DCS (Running, Fault) per room.	Noted	3
Air conditioner cooling capacity, rating and quantity	-		Vendor to complete **Table B** below.	2
Air conditioner fully installed at factory and tested	-	AC units to be fully installed at the factory (includes the outdoor unit, piping/electrical connections), tested during switchroom FAT, and to remain fully assembled for transport (preferred)	Noted where possible There may be a need site reconnection of the air conditioner gas pipe work & electrical connections. This will be determined during detailed design phase.	1
A/C outdoor unit installed for transport	-	Yes – outdoor units are installed on ends of Switchroom and transported attached to room. Vendor to advise for larger units.	Noted where possible There may be a need site reconnection of the air conditioner gas pipe work & electrical connections. This will be determined during detailed design phase.	1
Combined Temperature & Humidity Sensor	Yes / No	Yes – One per switching room and transformer room. Not required in airlock. 4-20mA output to DCS. Vendor to list make/model.	Yes	1
Pressurisation fan	Yes / No	No		1
LV Switchgear heat load	kW	VTA – **refer to Table A**	Vendor to complete Table A below.	2
VSD heat load	kW	Refer to VSD Heat loads in **Table A** below (to be confirmed after award)		1
Switchroom internal equipment heat load	kW	Refer to Note 11 & **Table A**. Equipment heat load includes 20% fouling/safety factor.	Vendor to complete **Table A** below.	2
Switchroom environmental heat load (Passive heat load)	kW	Refer to **Table A**	Vendor to complete **Table A** below.	2
Remote Switching Panels				
Design, supply and Install	Yes / No	Yes – HV Switchgear only	Yes	1
Control Voltage	-	48VDC	Noted	1
Indicator Light Type	-	Semaphore	Noted	1
Supply and Install TemRack IR LCS	Yes / No	Yes – One per LV Incomer ACB	Yes	1
Location	-	In airlock	Noted	1
Emergency Shutdown Pushbutton				
Design, supply and Install	Yes / No	Yes – One per switchroom	Yes	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Design	-	Red mushroom head latching push button comes with yellow ring, and a shroud or lift-up flap to prevent inadvertent operation. Clear warning signage required.	Noted	1
Contacts	-	2 N/O contacts per LV incomer 2 N/O contacts per HV incomer 1 N/O contact for DCS	Noted	1
Location	-	In airlock	Noted	1
MCC				
Floor Space for Future Expansion	-	Supplier shall provide floor space adequate for one future functional double sided tier and cable zone tier on the extendable side of the MCC	Noted and we comply. (Sub floor steel work is provided, however floor cutout and drop boxes have not be allowed for)	1
Arc Fault Detection – Current transformers	-	Supplier to install Current transformers on purposedly built frame in the Transformer Room.	Yes on a mounting frame off the transformer	1
Electrical Services				
Supply and installation of the below electrical services				
Interior Normal Lighting and Switches	Yes / No	Yes, Internal lighting and light switches	Yes	1
Emergency and Exit Lighting to AS2293 supplied and installed by vendor	Yes / No	Yes, Emergency and exit lighting to AS2293	Yes	1
External Wall Mounted Luminaires	Yes / No	Yes, LED lights mounted above each egress doorway. One External wall mounted LED flood light on each side of the building. Wall mount Emergency lighting to AS2293 required for all landing stairway.	Yes	1
Photoelectric cell	Yes / No	Yes	Yes	1
Small Power	Yes / No	Yes, Internal GPOs. A/C Isolators (one per A/C)	Yes	1
L&SP Distribution Board	Yes / No	Yes, 3 Phase L&SP DB to supply building services, with auto/manual selector for external lighting, photocell relay output to control auto external lighting. FIP to control contactors for A/C fire system trip. Minimum 20% spare capacity, and Rapid Test RCD Test system included Vendor to size DB to suit the number of Air Conditioning Circuits.	Yes	1
415V HV Auxiliary Supply Distribution Board	Yes / No	Yes, one located in F16-SR-001	Yes	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
415V or 240VAC UPS Distribution Board	Yes / No	Yes, 240VAC UPS Distribution Board for Communications, MCC control supply, DCS. 1x 415Vac UPS Distribution Board for F16-SR-002	Yes	1
Cabling for Mechanical, Electrical, Comms and Fire Services	Yes/No	Yes, Cabling for mechanical, electrical and fire services. Cabling and supports between all room equipment (including free-issued equipment) as required. For free issue equipment cabling, refer to the equipment list and cable schedule for each switchroom. Vendor is responsible for cable sizing of all cables between switchroom equipment.	Yes We have followed the EPC cable schedule for all cabling to the equipment. SL&P, HVAC, Fire cabling is contained withing the wall cavity and is included in our offer	1, 7, 8, 9
Communication Cable segregation	-	For all equipment mounted against the wall, comms cables are to be run in duct, mounted above the equipment, to communications cabinet. For equipment mounted in the centre of the room, there shall be a dedicated, segregated, 150mm cable ladder, mounted in the switchroom subframe, to the communications cabinet.	Noted. For subfloor we have allowed for a comm's cable trunk for all comms cables	1
Wall mount VSD Support	Yes/No	Yes, Vendor to design and install a purposedly built wall frame to support the free issued wall mount VSDs including spare space for future use.	Yes	1
Main Earth Bar	Yes/No	Yes, in Switchroom base frame. Pre-drilled and sized to suit all Switchroom internal connections, plus 12 external connections and 20% spare capacity:	Yes	1
Earthing Cables	Yes/No	Yes, Incl. Earthing of free-issued and supplied equipment and building structure.	Yes	1, 7
Fire Services and Safety				
Fire panel, manual buttons and sensors supplied and installed as shown on drawings	Yes / No	Yes		1
Fire Panel make	-	Wormald	Tyco	3
Fire Panel model	-	VTA	MX1 (addressable)	3
Design notes	-	N/B		1
Smoke and heat detector make and model	-	VTA	Photo Optical	3
Aspirating Smoke Detector (ASD) make and model	-	Xtralis VESDA VEP-A00-1P	VES-A00-P or VEP-A00-P	3

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
Aspirating Smoke Detector (ASD) sample points	-	1 pipe with LED's Aspirating System. Multiple sampling points, Transformer enclosures, aspirating tube along Ceilings, LV compartments of HV switchgear, LV MCC cable zones, LV VSDs >315kW, DBs >100Amps, DCS cabinets, RIO panels, UPS cabinet, quantity TBC during detailed design.	Noted (see technical offer for details) 1 pipe not sufficient 4 pipe allowed for	3
Manual glass break type alarm buttons make and model	-	VTA	Gas local control call point	3
Fire panel input voltage	-	240VAC 50Hz single phase	Yes	1
Backup battery	Yes / No	Yes	Yes	1
Fire Panel Communications	-	2 fibre optic modems with ST connectors, 2 SMOF outputs each, connection to central fire detection and alarm system.	Yes	1
Spare relay contact (for status/alarm to DCS)	Yes / No	Yes – Alarm, Isolate, Fault	Yes	1
Automatic trip to air-conditioners	Yes / No	Yes	Yes	1
Gas suppression systems – General	Yes / No -	Yes, incl. Inergen (or equivalent) gas cylinder/s, valves, piping and cabinets; Cylinder/s to be installed indoor within the Switchroom.	Yes	1
Gas suppression systems • Cylinder Gas type	-		IG541 (Frike Pro Inert)	2
Gas suppression systems • Cylinder Qty	-		As commercial offer	2
Gas suppression systems • Cylinder Weight and dimension (Each)	-		As commercial offer	2
Fire Extinguisher type	-	5kg dry powder (internal & external)	5KG CO2 AS PER AS 2444	1
Fire extinguisher wall mount brackets and signs provided	Yes / No	Yes	Yes	1
Fire blankets provided	Yes / No	Yes, 2 off 1.2 x 1.8m	Yes	1
Fire blankets mounted on walls	Yes / No	Yes	Yes	1
Fire aid kit and LV resuscitation kit supplied and installed as shown on drawings	Yes / No	Yes	Yes	1
Safety signage and resuscitation chart supplied and installed	Yes / No	Yes	Yes	1
HV Rescue Kit provided	Yes / No	Yes – in each HV switchroom	Yes	1
LV Rescue Kit provided	Yes / No	Yes	Yes	1
Automatic external defibrillator	-	Lifepak Defibrillator CR2 Fully Automatic	Yes	3
Equipment within Switchroom				
Supply and installation of equipment and cabling	-	As per the equipment list and cable schedule of each switchroom	Yes	1
Labelling				
Switchroom Entrance Labels:				

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
• Access doors		Yes; Switchroom Identification label, danger sign, voltage warning sign	Yes	1
• Transformer Room Access – Additional Label		Yes, Isolate before entry label, high voltage warning sign	Yes	1
Arc Flash Labels	Yes / No	Yes, EPC to supply Label data, Vendor to supply and install labels. Labels to be fitted on all HV switchgear, HV VSDs, LV MCCs and >100A rated LV DB/VSDs	Yes	1
Traffolyte label required on switchroom equipment (including free issue equipment supplied without labels)	Yes / No	Yes, etched traffolyte label (black on white background) on switchroom equipment.	Yes	1
Label Information	-	Equipment number, equipment description, and schematic drawing number.	Yes	1
"Danger xxxV" labelling on equipment supplied with voltages above ELV	-	Yes, traffolyte label. Traffolyte "Fed from" label with supplied from equipment/circuit also required.	Yes	1
Emergency Shutdown Pushbutton Label		Labelled with" "EMERGENCY SWITCHROOM SHUTDOWN – ACTIVATING THIS PUSHBUTTON WILL TRIP THE SWITCHBOARD INCOMING POWER SUPPLY""	Yes	1
Lighting and Outlets	-	Labelled with circuit number.	Yes	1
Test Requirements				
Tests Required:				
Inspection	-	Yes	Yes	1
Routine	-	Yes	Yes	1
Special	-	Yes, including L&SP, Rapidtest system, Air conditioning system and Fire systems (inclusive of room air tightness integrity test)	Yes	1
Commissioning	-	Full off-site commissioning of work completed by vendor. Allowance for 5 days of integration testing to be performed by EPC at vendor's premises (per Switchroom). Temporary power to be available to switchboard for the duration of the integration testing.	Noted	1
Shipping and Delivery				
Overall Dimensions of Building	L x W x H (mm)	Vendor to complete below **Table C** with switchroom dimensions. Length, Width and Height (for shippings and internal ceiling height	Vendor to complete **Table C** below.	2
Total Weight of Building	kg	Vendor to complete below **Table C** with approximate shipping weights for tender. Includes building, attached platforms & installed equipment	Vendor to complete **Table C** below.	2
All penetrations sealed before shipping	Yes / No	Yes	Yes	1
All loose items, such as external light fittings and fire extinguishers are packed into switchroom before shipping	Yes / No	Yes	Yes	1

DUTY SPECIFICATION	UOM	EPC DATA VTA = Vendor to Advise	SUPPLIER DATA Supplier to Complete blank cells and indicate acceptance or otherwise as appropriate	NOTES
All batteries to be demounted and packed into switchroom for transport.	Yes / No	Yes	Yes	1
Temporary Bracing provided for any equipment requiring additional support before shipping provided	Yes / No	Yes	Yes	1

Notes:

1. Data supplied by EPC The Supplier shall comply with this data, as a minimum. Supplier shall state their applicable ratings/data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. Equipment layout drawing shown is indicative. Supplier shall provide room layout complete with dimensions, including penetrations and subfloor cable ladder. Preliminary Layouts to be provided with submission of tender.
5. Loading diagrams shall be provided with documentation, in accordance with note 6.
6. Loads at the loading points shall be provided as unfactored and for primary load cases (Self weight, Live load, Wind etc) and shall be in accordance with AS 1170. Designated loading points shall be specified by supplier.
7. Power, control and earthing cables between equipment installed in the room to be run on cable ladder installed within the switchroom baseframe. Supply of this cabling and cable ladder to be by the Supplier.
8. Internal room lighting and small power cables to be run in duct/conduit internal to the room.
9. Communications block diagrams shall be provided post award. Refer to each switchrooms equipment list and cable schedule for indicative patch leads required.
10. HVAC sizing calculations to be performed by switchroom supplier confirming size, selection and capacity of HVAC systems for given site conditions, equipment heat loads and passive heat gain.
11. HVAC design to confirm H^2 build up from battery charging (415Vac UPS and 48V DC battery) meets AS2676.2 ventilation requirements.

Table A Equipment Heat Load

Vendor to complete below table.

NO.	SWITCHROOM	VSD HEAT LOAD (KW) From load list	HARMONIC FILTER HEAT LOAD (KW)	DRY TRANSFORMER HEAT LOAD (KW) PER ENCLOSURE	MCC & MISC. EQUIPMENT HEAT LOAD (KW) Supplier To Complete Blank Cells	ENVIRONMENTAL HEAT LOADS –SOLAR RADIATION + PASSIVE HEAT LOAD (KW) Supplier To Complete Blank Cells	TOTAL BUILDING HEAT LOAD (KW)
1	F00-SR-001	Not applicable	Not applicable	Not applicable	20	30KW	54KW
2	F12-SR-001	22.82	Not applicable	Not applicable	5	7.7KW	41.7KW
3	F12-SR-002	16.6	Not applicable	16	9.5	17.6KW	68.1KW
4	F13-SR-001	1	Not applicable	18	11	18.1KW	55KW
5	F13-SR-002	12.14	Not applicable	18	10.25	18KW	67KW
6	F13-SR-003	28	Not applicable	Not applicable	5	30KW	70KW
7	F13-SR-004	28.42	7.1	18	9	19.7KW	95KW
8	F13-SR-005	0.41	7.1	18	9	18KW	60KW
9	F16-SR-001	177.46	Not applicable	Not applicable	15	38.5KW	270KW
10	F16-SR-002	52.33	7.1	18 / 18 / 5.5	21.5	40.1KW	187.8KW
11	F21-SR-001	9.45	Not applicable	18 / 18	15	31KW	104.3KW
12	F22-SR-001	13	Not applicable	18 / 18	15	31KW	95KW
13	F23-SR-001	75.4	Not applicable	Not applicable	7	16.5KW	134.5KW
14	F23-SR-002	0.87	Not applicable	10	6	12.1KW	32.3KW
15	F28-SR-001	0	Not applicable	Not applicable	5	6.6KW	12.6KW
16	F28-SR-002	6.88	Not applicable	10	9	16.5KW	57.4KW
17	F72-SR-001	14.1	Not applicable	12			

Table B HVAC Cooling Capacity

Vendor to complete below table.

NO.	SP.	SWITCHROOM	UNIT QTY (DUTY)	UNIT QTY (Stby)	SENSIBLE COOLING (kW) (Duty)	SENSIBLE COOLING (kW) (Standby)	TOTAL SENSIBLE COOLING (kW)
1	SP1	F00-SR-001	4	1	50	12.5	62.5
2	SP1	F12-SR-001	3	1	42	14	56
3	SP1	F12-SR-002	4	1	56	14	70
	SP1	F12-TR-001	2	1	24	14	42
4	SP1	F13-SR-001	3	1	42	14	56
	SP1	F13-TR-002	2	1	24	14	42
5	SP1	F13-SR-002	3	1	42	14	56
	SP1	F13-TR-003	2	1	24	14	42
6	SP1	F13-SR-003	5	1	70	14	84
7	SP1	F13-SR-004	5	1	70	14	84
	SP1	F13-TR-014	2	1	24	14	42
8	SP1	F13-SR-005	3	1	42	14	56
	SP1	F13-TR-020	2	1	24	14	42
9	SP1	F16-SR-001	20	1	280	14	294
10	SP1	F16-SR-002	10	1	140	14	154
	SP1	F16-TR-004	2	1	24	14	42
	SP1	F16-TR-005	2	1	24	14	42
	SP1	F16-TR-008	1	1	10	10	20
11	SP2	F21-SR-001	4	1	56	14	70
	SP2	F21-TR-006	2	1	24	14	42
	SP2	F21-TR-007	2	1	24	14	42
12	SP2	F22-SR-001	4	1	56	14	70
	SP2	F22-TR-030	2	1	24	14	42
	SP2	F22-TR-031	2	1	24	14	42
13	SP1	F23-SR-001	10	1	140	14	154
14	SP1	F23-SR-002	2	1	20	10	30
	SP1	F23-TR-023	2	1	20	10	30

15	SP1	F28-SR-001	2	1	14.2	7.1	21.3
16	SP1	F28-SR-002	3	1	37.5	12.5	50
	SP1	F28-TR-009	2	1	28	14	42
17	SP2	F72-SR-001					
		F72-TR-072					

Table C Switchrooms Shipping Dimensions and Weights

Vendor to complete below table*

NO.	SWITCHROOM/S	LENGTH (mm)	WIDTH (mm)	OVERALL HEIGHT (mm)	INTERNAL CEILING HEIGHT (mm)	TOTAL SHIPPING WEIGHT (kg)
1	F00-SR-001	27,000	5000	TBC	Min 3500	TBC
2	F12-SR-001 & 002	25000	5000	TBC	Min 3500	TBC
3	F13-SR-001	18500	5000	TBC	Min 3000	TBC
4	F13-SR-002	18500	5000	TBC	Min 3000	TBC
5	F13-SR-003	21000	6500	TBC	Min 3500	TBC
6	F13-SR-004	20000	5000	TBC	Min 3000	TBC
7	F13-SR-005	18000	5000	TBC	Min 3000	TBC
8	F16-SR-001	35000	5000	TBC	Min 3500	TBC
9	F16-SR-002	35000	5000	TBC	Min 3000	TBC
10	F21-SR-001 (SP2)	28000 (31,000 extended)	5000	TBC	Min 3000	TBC
11	F22-SR-001 (SP2)	30000	5000	TBC	Min 3000	TBC
12	F23-SR-001 & 002	28000	5000	TBC	Min 3500	TBC
13	F28-SR-001 & 002	23000	5000	TBC	Min 3500	TBC
14	F72-SR-001 (SP2)	17000	5000	TBC	Min 3000	TBC

*Vendor to specify where multiple shipping sections are required for a particular building.